

Disease Detectives C - Disease Detectives - Rickards Invitational Div. C SATELLITE - 11-01-2025
Student Answer Sheet (/rickards/ES/AnswerSheet?tid=000GJ1)

Instructions (shown before students start the test)

Have fun! Follow nats rules.

Introduction (shown after students start the test)

Hello! You will have 50 minutes to complete the following test. **ANY** out-of-browser time will be recorded and may result in penalties.



1. (1.00 pts) Which of the following best represents a secular trend?

- ☐ A) Declining syphilis rates after antibiotics were introduced in the 1940s
- ☐ B) Rising obesity prevalence across all U.S. adult age groups from 1980 to 2020
- ☐ C) Higher gastric cancer risk in people born before refrigeration was common
- ☐ D) Increased lung cancer only among men born in the 1930s who smoked

2. (1.00 pts) In a natural history of disease framework, the period after exposure but before clinical symptoms is known as:

- ☐ A) Incubation period
- ☐ B) Latency period
- ☐ C) Prodromal period
- ☐ D) Subclinical disease period

3. (1.00 pts) Which of the following biases is most likely when hospital controls are used in a case-control study?

- ☐ A) Recall bias
- ☐ B) Berkson's bias
- ☐ C) Selection bias due to non-response
- ☐ D) Surveillance bias

4. (1.00 pts) A screening test that identifies true negatives very accurately but misses many true positives has:

- ☐ A) High sensitivity, high specificity
- ☐ B) High sensitivity, low specificity
- ☐ C) Low sensitivity, high specificity
- ☐ D) Low sensitivity, low specificity

5. (1.00 pts) Which of the following scenarios demonstrates differential misclassification?

- ☐ A) Both cases and controls inaccurately recall their past diet
- ☐ B) Only cases with a disease are more likely to misreport past exposures
- ☐ C) Laboratory test occasionally fails to detect an infection equally in both groups
- ☐ D) Random undercounting of cases during surveillance

6. (1.00 pts) A sentinel surveillance system is most useful for:

- ☐ A) Measuring the exact incidence of a disease
- ☐ B) Detecting rare adverse events after vaccine rollout
- ☐ C) Rapid outbreak detection with full population coverage

- ☐ D) Capturing all cases of notifiable diseases

7. (1.00 pts) Which study design allows a participant to serve as their own control?

- ☐ A) Case-control study
- ☐ B) Case-crossover study
- ☐ C) Prospective cohort study
- ☐ D) Ecological study

8. (1.00 pts) Which Bradford Hill criterion is best supported when multiple studies in different populations consistently find the same exposure-disease relationship?

- ☐ A) Specificity
- ☐ B) Consistency
- ☐ C) Temporality
- ☐ D) Coherence

9. (1.00 pts) Which of the following is an example of a continuous common source outbreak?

- ☐ A) Multiple norovirus cases after a cruise ship dinner
- ☐ B) Cholera from a contaminated city water supply over several weeks
- ☐ C) Influenza spreading person-to-person at a school
- ☐ D) Lead poisoning from a one-time community exposure

10. (1.00 pts) Which prevention level is illustrated by HIV antiretroviral therapy to reduce viral load and slow progression of disease?

- ☐ A) Primary
- ☐ B) Secondary
- ☐ C) Tertiary
- ☐ D) Quaternary

11. (1.00 pts) The “healthy worker effect” usually causes:

- ☐ A) Overestimation of occupational risk
- ☐ B) Underestimation of occupational risk
- ☐ C) Equal estimation of risk compared to the general population
- ☐ D) Overmatching in occupational studies

12. (1.00 pts) Which concept is best represented by the iceberg phenomenon?

- ☐ A) Spectrum of disease
- ☐ B) Chain of infection
- ☐ C) Reservoir dynamics
- ☐ D) Agent-host-environment interaction

13. (1.00 pts) Which case control study design tries to get rid of confounding without randomization?

- ☐ A) Nested case-control
- ☐ B) Case cohort
- ☐ C) Matched case control
- ☐ D) Case crossover

14. (1.00 pts) The time between exposure and first infectiousness (not symptoms) is called:

- ☐ A) Incubation period

- ☐ B) Latent period
- ☐ C) Serial interval
- ☐ D) Generation time

15. (1.00 pts) Which of the following is not a core step in an outbreak investigation?

- ☐ A) Verify diagnosis
- ☐ B) Construct the epidemic curve
- ☐ C) Develop policy legislation
- ☐ D) Establish case definition

16. (1.00 pts) Which bias occurs if patients who know their exposure status are more likely to report illness symptoms?

- ☐ A) Observer bias
- ☐ B) Neyman's Bias
- ☐ C) Surveillance bias
- ☐ D) Information bias

17. (1.00 pts) Which of the following public health agencies is not typically considered a direct public health authority in the U.S.?

- ☐ A) CDC
- ☐ B) FDA
- ☐ C) FTC
- ☐ D) NIH

18. (1.00 pts) Which concept is illustrated when a diagnostic test is performed in a high-prevalence setting, resulting in a higher positive predictive value?

- ☐ A) Spectrum bias

- ☐ B) Bayes' theorem
- ☐ C) Overdiagnosis bias
- ☐ D) Detection bias

19. (1.00 pts) Which of the following is least aligned with the definition of One Health?

- ☐ A) Human-animal-environment connections
- ☐ B) Vector control to reduce malaria transmission
- ☐ C) Economic incentives for pharmaceutical development
- ☐ D) Reducing antimicrobial resistance in livestock

20. (1.00 pts) In surveillance, the process of "data interpretation" comes immediately after:

- ☐ A) Data dissemination
- ☐ B) Data collection
- ☐ C) Link to action
- ☐ D) Hypothesis testing

21. (1.00 pts) Which of the following represents overmatching in a case-control study of foodborne illness?

- ☐ A) Matching cases and controls on age to reduce confounding
- ☐ B) Matching on food consumption
- ☐ C) Randomly selecting community controls
- ☐ D) Matching on demographic characteristics

22. (2.00 pts) Define lead time bias and provide an example involving an infectious disease.

23. (3.00 pts) Compare and contrast passive and active surveillance in terms of cost, coverage, and timeliness. Give one situation where each would be optimal.

24. (3.00 pts)

The concept of the spectrum of disease explains differences in outcomes for the same pathogen. Provide one example each for a mild, subclinical, and severe presentation of a single disease.

25. (2.00 pts) Explain how misclassification of exposure can bias results differently if it is non-differential vs differential.

26. (2.00 pts) Describe the iceberg phenomenon using hepatitis B as an example. What does this imply for control measures?

27. (2.00 pts) Using the epidemiologic triad, explain how vector control affects malaria transmission differently from human prophylaxis.

28. (1.00 pts) A researcher designs a case-control study for lung cancer but matches controls to cases on smoking status. Why is this a methodological flaw?

29. (3.00 pts) Define confounding and describe one strategy to control for it in both the design and analysis phases of a study.

30. (2.00 pts) Explain how syndromic surveillance could detect an outbreak of a novel respiratory pathogen earlier than laboratory-based surveillance. What are its tradeoffs?

31. (13.00 pts) List all 13 Steps of Outbreak Investigation

32. (1.00 pts) What is the Hippocratic Oath

U.S. Air Force basic training takes place at Joint Base San Antonio-Lackland in San Antonio, Texas. Trainees are organized into rolling military units averaging 900 entering and graduating trainees per week, with each unit further divided into groups of approximately 52 trainees. Trainees are assigned to specific dormitories, which typically house 50–60 persons, sleeping in beds that alternate head directions, creating a foot-to-head orientation in the bay. Trainees arrive weekly and undergo a standardized 7.5-week basic military training (BMT) curriculum organized by week of training, resulting in training activities and associated exposures that are typically consistent for each class.

To prevent invasive meningococcal and streptococcal disease outbreaks, all trainees receive quadrivalent meningococcal (Menveo) vaccine within 72 hours of arrival and a single dose of penicillin G benzathine injectable suspension, respectively, within 7 days of arrival (1). Penicillin allergic trainees receive weekly oral azithromycin to prevent streptococcal disease during BMT.

In February 2025, two cases of *Neisseria meningitidis* bacterial conjunctivitis were identified in otherwise healthy basic military trainees at Joint Base San Antonio-Lackland in San Antonio, Texas; an investigation was conducted to identify the source of the outbreak and to make treatment recommendations. Ultimately, 41 cases of *N. meningitidis* conjunctivitis and 32 cases of *Haemophilus* species conjunctivitis were identified among 11,797 trainees.

People spread meningococcal bacteria to others by sharing respiratory and throat secretions (saliva or spit). Generally, it takes close or lengthy contact to spread the bacteria.

The 41 confirmed cases of *N. meningitidis* conjunctivitis occurred in trainees in 37 unique BMT groups. During this period, men constituted 78% of the BMT population but accounted for 90% of the *N. meningitidis* cases. Among trainees with confirmed *N. meningitidis* conjunctivitis, 33 (80%) reported an antecedent upper respiratory infection, 27 people had cough, and 22 people had a sore throat. Overall, 35 (85%) patients had unilateral eye involvement.

Assume all confirmed cases are actual cases.

33. (1.00 pts) Give a possible reason why men are 90% of the cases, but 78% of the population, with no assumptions on meningitis.

34. (1.50 pts) Give three ways in which this population is not representative of the US.

35. (2.00 pts) Give a case definition for this outbreak.

36. (2.00 pts) Should case definitions be sensitive or specific? Why?

37. (2.00 pts) Assume 1/47 of all trainees have a cough. Calculate the sensitivity and specificity of the case definition, "a trainee within the time period who has a cough".

38. (1.00 pts) Why doesn't everyone who is confirmed have the antecedent upper respiratory infection?

39. (2.00 pts) What level(s) of prevention are conducted within the military base?

"...Penicillin allergic trainees receive weekly oral azithromycin to prevent streptococcal disease during BMT"

40. (2.00 pts) Name the above procedure, and give an example of another situation where it is used.

41. (2.00 pts) If possible, calculate the effectiveness (not efficacy) of the Menveo vaccine. If not, explain why.

42. (2.00 pts) Give a risk factor for the soldiers, as presented in the article, and give a suggestion on how to mitigate it reasonably.

43. (2.00 pts) Give a way to implement your suggestion that would not be costly.

For questions 44-47, assume that everyone has an equal probability as everyone else of contracting disease given an exposure, and everybody has an equal level of exposure to the disease. Assume further the disease has no incubation period.

44. (2.00 pts) What is the SIR model? Explain what S, I, and R stand for, and explain the parameters needed for the model.

45. (2.50 pts) Give five limitations of the SIR model.

46. (1.00 pts) The SIR model is a “compartmental” model. Explain what this means.

47. (2.00 pts) Can the SIR model be used to completely model this outbreak? If not, why not? (we will not make you calculate the parameters)

The following is pulled from the same case study.

“Whole genome sequencing of isolates from the first two cases suggested that the organism was unencapsulated (nongroupable) and that the cases were related.”

48. (2.00 pts) How do we copy DNA in order to do whole genome sequencing? Give some detail into this process.

49. (2.00 pts) Why is the step above needed?

50. (2.00 pts) How does the use of whole genome sequencing here align with the idea of PulseNet?

On February 5, 2025, a case of *N. meningitidis* bacterial conjunctivitis was identified in an otherwise healthy BMT trainee who had experienced 2 days of mucopurulent ocular discharge; the trainee had no known exposure to *N. meningitidis*. Although the symptoms initially suggested viral conjunctivitis, the copious unilateral discharge led the health care provider to culture the exudate, which was positive for *N. meningitidis* 6 days later.

Microbial isolates were tested and identified by Vitek 2 (bioMérieux). On February 21, a second trainee (from a different training unit) was also evaluated for unilateral copious mucopurulent ocular discharge associated with periocular edema without corneal involvement.

The culture of the exudate from this trainee's eye was positive for *N. meningitidis*, raising concern about potential cases of invasive meningococcal disease. An investigation was initiated to identify additional cases, determine the risk for invasive meningococcal disease, and describe the patients and evaluate their response to treatment.

51. (1.00 pts) Name a step of outbreak investigation performed above.

A (completely made up) table of results is shown below. Assume the Gold Standard has a 97% PPV and 98% NPV.

	Gold Standard Positive	Gold Standard Negative
Videk 2 Positive	900	30
Videk 2 Negative	100	470

52. (1.00 pts) Give a difference between the pairs of sensitivity / specificity and PPV / NPV. Define and use the term “Intrinsic Property” in your answer.

53. (1.00 pts) Give an example of a “good” test having a “bad” PPV / NPV.

54. (2.00 pts) Calculate the sensitivity and specificity of the gold standard.

55. (1.00 pts) Calculate the number of cases.

56. (1.00 pts)

Define series and parallel testing. What happens to the sensitivity and specificity of the result if we do two gold standard tests in parallel? What if we conduct it in series? (Do not calculate, just tell us the direction of change)

57. (2.00 pts) Why would the information we gave you be unrealistic to obtain in a real-world setting?**58. (3.00 pts)** Estimate the sens/spec/npv/ppv of Videk 2. Explain your estimates.**59. (3.00 pts)** What would be the sensitivity and specificity of if you did a gold standard test and then videk 2 test in series? Explain your answer.**60. (2.00 pts)** Does it make sense to do the above test?

Suppose that the epidemiologists trace back all the cases, and find out that many of the cases are linked back to a certain place. Suppose further that this place is my house. Consider the following table that I made up.

	Got Meningitis	Did not get meningitis
Went to my house	600	50
Did not go to my house	80	4

(My house is very spacious.)

61. (1.00 pts) Calculate an appropriate measure of association between getting meningitis and going to my house.

62. (1.00 pts) What type of study was performed?

63. (3.00 pts) Can you perform a Chi-Square test on these results?

If YES, calculate the appropriate statistic and come to an appropriate conclusion ($p=0.05$, critical value = 3.841). Name the appropriate Chi-Square test.

If NO, give an explanation as to why. Furthermore, name a test that could be used instead (do not calculate the statistic).

64. (1.00 pts) Who is credited with first shifting explanations of disease from supernatural causes to rational observation?

- ☐ A) Hippocrates
- ☐ B) Virchow
- ☐ C) Jenner
- ☐ D) Koch

65. (1.00 pts) John Graunt's analysis of mortality bills in London is significant because it:

- ☐ A) Introduced randomization in trials
- ☐ B) Created early population-level statistics

- ☐ C) Demonstrated germ theory
- ☐ D) Proved the effectiveness of inoculation

66. (1.00 pts) Who introduced smallpox inoculation to Britain after learning the practice in Turkey?

- ☐ A) Louis Pasteur
- ☐ B) Lady Mary Wortley Montagu
- ☐ C) Alexander Fleming
- ☐ D) Edward Jenner

67. (1.00 pts) James Lind's experiments on sailors suffering from scurvy were notable because they:

- ☐ A) Involved concurrently treated control groups
- ☐ B) Applied Koch's postulates to diet
- ☐ C) Used Gram staining
- ☐ D) Focused on mosquito-borne disease

68. (2.00 pts) Which french physician studied the black plague in the 16th century, recommending good hygiene and bodies be removed from the street?

69. (1.00 pts) In what year did the first official English translation of Graunt's Natural and Political Observations Made upon the Bills of Mortality appear?

- ☐ A) 1662
- ☐ B) 1665
- ☐ C) 1676
- ☐ D) 1681

70. (1.00 pts) What was the title of James Lind's published work where he described his scurvy experiments?

- ☐ A) *The Nature and Cure of the Scurvy*
- ☐ B) *Treatise of the Scurvy*
- ☐ C) *Observations on the Naval's Medicine*
- ☐ D) *Observations of the Scurvy*

71. (1.00 pts) Edward Jenner was elected to which scientific society in recognition of his vaccination research?

- ☐ A) The Royal Society
- ☐ B) The Royal College of Physicians
- ☐ C) The Linnean Society
- ☐ D) The Edinburgh Medical Society

There is an outbreak of Rickardsitis disease, and you have been assigned to investigate. To learn more about Rickardsitis, you look at past epidemiology studies on the disease. The data below is taken from a small town (population size 250) where there were 185 new cases of TQD over November. A pharmaceutical company called Invitationals Incorporated has developed a test to detect Rickardsitis. Here are the results of the evaluation study.

	Invitationals Inc Positive	Invitationals Inc Negative
Tested positive	125	60
Tested negative	20	45

Using the table calculate the following values to the nearest hundredth.

72. (0.50 pts) What is the Sensitivity of the Invitationals Inc. Test?

73. (0.50 pts) What is the Specificity of the Invitationals Inc. Test?

74. (0.50 pts) What is the PPV of the Invitationals Inc. Test?

75. (0.50 pts) What is the NPV of the Invitationals Inc. Test?

76. (2.00 pts) Should the CDC use the test?

77. (1.00 pts) How would one aim to improve the sensitivity of the test?

Huzzah! Invitationals Inc. has also crafted a vaccine for Rickardsitis, to ensure that it works they ran a trial with some getting a vaccine dose others getting a placebo with the results below

	Got Disease	Did not get disease
Took Vaccine	20	105
Did Not Take	45	85

78. (1.00 pts) What type of immunity are Vaccines?

- ☐ A) Active Immunity
- ☐ B) Passive Immunity

79. (1.00 pts) What about the type of Prevention?

- ☐ A) Primary
- ☐ B) Secondary
- ☐ C) Tertiary
- ☐ D) Quaternary

80. (2.00 pts) Calculate the Vaccine Efficiency.

For Rickardsitis, the transmissivity is .5, amount of people contacted per day on average is 25, and an individual on average is infected for 3 days.

81. (1.00 pts) Given this Information calculate R0

82. (2.00 pts) Using your calculated values calculate and interpret the Herd Immunity Threshold

The following case study is completely made up, any resemblance to real life figures or events is coincidental.

Dr. Imogen Vale, a senior field epidemiologist at the Department of Environmental Health, was called to investigate an unusual rise in chronic fatigue and skin irritation among urban gardeners in Brookmere City. The condition, later termed Blight Fever, presented with mild fevers, weakness, and a faintly luminescent rash visible under UV light. Concern escalated when local physicians noticed a pattern: nearly all affected gardeners worked in one of three community greenhouses that had recently switched to a new soil additive called EcoBloom X, designed to improve soil nitrogen retention.

To determine whether EcoBloom X was linked to Blight Fever, Dr. Vale and her team identified all active gardeners in the three greenhouses and recorded whether they had developed Blight Fever within three months of EcoBloom X's introduction. They also measured each gardener's level of exposure to the additive, categorizing them as high, medium, or low exposure based on hours spent handling treated soil. A blood test was later developed to detect a biomarker called BF- α , suspected to be associated with the illness.

Dr. Vale's team followed gardeners, observing who developed Blight Fever over time while documenting their baseline exposures and environmental factors such as greenhouse ventilation and humidity. Participants were contacted biweekly for symptom monitoring and lab testing. While this design provided valuable incidence data, the project required significant logistical coordination and funding, and many participants dropped out midway, citing time constraints and fatigue.

Here are two tables you will need

Table 1: Ecobloom x Exposure and Incidence of Blight Fever

Exposure Level	Blight Fever Cases	Non Cases	Total
High Exposure	90	110	200
Medium Exposure	55	165	220
Low Exposure	12	168	180

Table 2: BF- α Diagnostic Test Results Compared to Clinical Diagnosis

Test Result	Blight Fever +	Blight Fever -	Total
Positive	82	18	100
Negative	28	472	500
Total	110	490	600

83. (3.00 pts) Create a case definition for blight fever

84. (4.00 pts)

What type of epidemiological study is this (be specific)? Explain why and provide at least one advantage and one disadvantage of using this study design for an outbreak like this.

85. (3.00 pts) Using table 1 calculate the attack rates for the following

- a. High Exposure
- b. Low Exposure
- c. Medium Exposure

86. (2.00 pts) Calculate Relative Risk for the following and interpret its result

- a. High v. Low Exposure
- b. Medium v. Low Exposure

87. (1.00 pts) Compute the absolute risk difference (ARD) between the high and low exposure groups.

88. (3.00 pts) Construct a 2x2 table for high exposure vs. low exposure and calculate the odds ratio (OR). Compare it to your RR result and explain why the two values may differ. (For scilympiad purposes you can format answer as A (top left) : #, B (top right) : #, etc or format it as shown below)

a b

c d

89. (2.00 pts) Calculate the following using Table 2

Sensitivity

Specificity

PPV

NPV

90. (2.00 pts) Recalculate PPV and NPV assuming true disease prevalence of 5% and 20%. Explain how changes in prevalence affect the interpretation of a test's usefulness.

91. (2.00 pts) Compute Likelihood Ratios(LR), LR+ and LR- for the BF- α test and interpret what each value implies about the diagnostic power of the test.

The team later found that greenhouses with better ventilation used more EcoBloom X and also had lower humidity, which independently reduces disease risk.

92. (1.00 pts) Identify whether ventilation is a confounder, mediator, or collider.

93. (2.00 pts) Explain how failing to adjust for it could bias the results.

94. (1.50 pts)
If clinical diagnosis sensitivity = 0.9 and specificity = 0.95, adjust your observed attack rates for misclassification and recompute the corrected RR for high vs. low exposure.

95. (6.00 pts) Discuss potential sources of selection bias, information bias, and loss to follow-up bias in this investigation. Suggest how each could be minimized.

96. (1.00 pts) Would an active or passive surveillance system be better suited to monitor Blight Fever cases in Brookmere moving forward?

97. (2.00 pts) Name one advantage and one disadvantage to your chosen surveillance system (no points will be awarded if the answer for the previous question is incorrect)

Congrats on finishing the test, and good luck in your other events!

Feel free to share your thoughts about the test through this form: <https://forms.gle/NEyVbXdvz7Zdqsun6>